CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in machine learning and data analysis with Python, from feature engineering and predictive models to model validation and automation of analysis pipelines. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by predictive time-series modelling and a strong foundation in databases across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction pipeline end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) for complex regulatory and technical documents, validated against an SAP-system-derived ground truth; precision/recall raised from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for fast, high-quality development grounded in clean code principles.
- **Tech stack: Python, Neo4j, MongoDB, Postgres, LangGraph, LangChain, LiteLLM, GraphRAG, MCP, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application using LangGraph to optimize decision-making workflows for vector database queries, general inference and summarization tasks.
- Developed a React front-end and a Python backend with FastAPI, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, FastAPI, LangGraph, LangChain, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated generation and retrieval evaluation of a RAG system with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on large datasets (over 1M data chunks of Covestro's internal chemical data).

- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing and fine-tuning models for optimized retrieval.
- **Tech stack: Python, Scikit-Learn, PyTorch, TensorFlow, Hugging Face Transformers, LangChain, RAGAS, PytrecEval, AWS (S3, Lambda, SQS, DynamoDB, CDK), Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for time-series analysis of EV load.
- Developed an LSTM model that achieved 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Payment Fraud Detection Pipeline | PySpark | XGBoost | SHAP | SQL

- Predictive ML pipeline on the public synthetic PaySim payment dataset: PySpark feature engineering (velocity, rolling amount and counterparty features), XGBoost vs. Isolation Forest, compared to a rule-based baseline on precision/recall at a fixed alert budget.
- Explainable alerts with SHAP reasons per flagged transaction, fuzzy name screening against the public EU sanctions list, SQL feature table in Postgres and a model card.

Project: Life Expectancy Analysis | Regression | Feature Selection

- Regression models, visualization and feature selection to analyse drivers of life expectancy.

TECHNICAL SKILLSET

- **Data Science & ML frameworks:** Python, Scikit-Learn, XGBoost, PyTorch, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, time-series models (ARIMA, SARIMA, Prophet, LSTM)
- **Data & Databases:** SQL, Postgres, MongoDB, Neo4j, DynamoDB, SPARQL, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Engineering:** FastAPI, Python microservices, Docker, Kubernetes, Git, GitLab CI/CD, Streamlit, React (beginner)
- **Cloud & GenAI:** AWS (S3, EC2, Lambda, SQS, CDK), Azure, LangGraph, LangChain, Hugging Face Transformers, RAG
- **Languages:** German (B1), English (Business Fluent), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in CSE) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT